# Packet Tracer - Modify Single-Area OSPFv2

Cisco Networking Academy Lab Manual & Answer Key

**Instructor Note:** Red font color or gray highlights indicate text that appears in the instructor copy only.

## Addressing Table

| DEVICE | INTERFACE | IPV4 ADDRESS | SUBNET MASK | DEFAULT GATEWAY |
|---|---|---|---|---|
| **R1** | G0/0 | 172.16.1.1 | 255.255.255.0 | N/A |
| **R1** | S0/0/0 | 172.16.3.1 | 255.255.255.252 | N/A |
| **R1** | S0/0/1 | 192.168.10.5 | 255.255.255.252 | N/A |
| **R2** | G0/0 | 172.16.2.1 | 255.255.255.0 | N/A |
| **R2** | S0/0/0 | 172.16.3.2 | 255.255.255.252 | N/A |
| **R2** | S0/0/1 | 192.168.10.9 | 255.255.255.252 | N/A |
| **R2** | S0/1/0 | 209.165.200.225 | 255.255.255.224 | N/A |
| **R3** | G0/0 | 192.168.1.1 | 255.255.255.0 | N/A |
| **R3** | S0/0/0 | 192.168.10.6 | 255.255.255.252 | N/A |
| **R3** | S0/0/1 | 192.168.10.10 | 255.255.255.252 | N/A |
| **PC1** | NIC | 172.16.1.2 | 255.255.255.0 | 172.16.1.1 |
| **PC2** | NIC | 172.16.2.2 | 255.255.255.0 | 172.16.2.1 |
| **PC3** | NIC | 192.168.1.2 | 255.255.255.0 | 192.168.1.1 |
| **Web Server** | NIC | 64.100.1.2 | 255.255.255.0 | 64.100.1.1 |

## Objectives

- **Part 1:** Modify OSPF Default Settings
- **Part 2:** Verify Connectivity

## Scenario

In this activity, OSPF is already configured and all end devices currently have full connectivity. You will modify the default OSPF routing configurations by changing the hello and dead timers and adjusting the bandwidth of a link. Then you will verify that full connectivity is restored for all end devices.

## Instructions

### Part 1: Modify OSPF Default Settings

**Step 1: Test connectivity between all end devices.**

Before modifying the OSPF settings, verify that all PCs can ping the web server and each other.

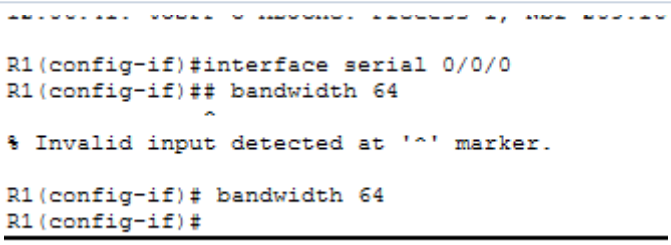
**Step 2: Adjust the hello and dead timers between R1 and R2.**

**a.** Enter the following commands on R1:

```
R1(config)# interface s0/0/0
R1(config-if)# ip ospf hello-interval 15
R1(config-if)# ip ospf dead-interval 60
```

**b.** After a short period of time, the OSPF connection with R2 will fail, as shown in the router output:

```
00:02:40: %OSPF-5-ADJCHG: Process 1, Nbr 209.165.200.225 on Serial0/0/0 from FULL to DOWN,
Neighbor Down: Dead timer expired
00:02:40: %OSPF-5-ADJCHG: Process 1, Nbr 209.165.200.225 on Serial0/0/0 from FULL to DOWN,
Neighbor Down: Interface down or detached
```




**รูปที่ 1: การตั้งค่า OSPF Timers บน Router R1 และ Log แสดงผลการขาดการเชื่อมต่อ (Neighbor Down)**

Both sides of the connection need to have the same timer values in order for the adjacency to be maintained. Identify the interface on R2 that is connected to R1. Adjust the timers on the R2 interface to match the settings on R1.

```
R2(config)# interface s0/0/0
R2(config-if)# ip ospf hello-interval 15
R2(config-if)# ip ospf dead-interval 60
```

After a brief period of time you should see a status message that indicates that the OSPF adjacency has been reestablished.

```
00:21:52: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.10.5 on Serial0/0/0 from LOADING to FULL,
Loading Done
```

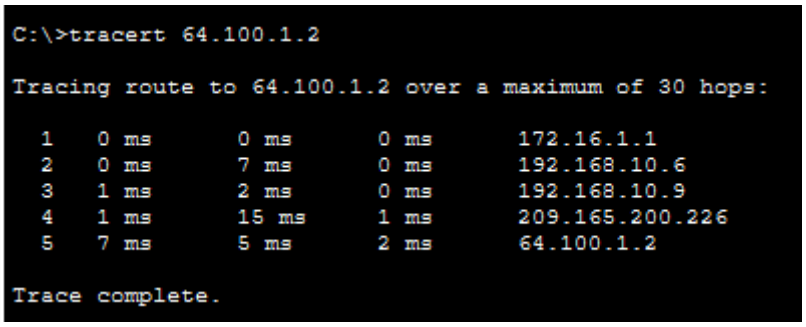
**Step 3: Adjust the bandwidth setting on R1.**

**a.** Trace the path between PC1 and the web server located at `64.100.1.2`. Notice that the path from PC1 to `64.100.1.2` is routed through R2. OSPF prefers the lower cost path.

```
C:\> tracert 64.100.1.2
Tracing route to 64.100.1.2 over a maximum of 30 hops:
1   1 ms   0 ms   8 ms   172.16.1.1
2   0 ms   1 ms   0 ms   172.16.3.2
3   1 ms   9 ms   2 ms   209.165.200.226
4   *      1 ms   0 ms   64.100.1.2
Trace complete.
```

**b.** On the R1 Serial 0/0/0 interface, set the bandwidth to 64 Kb/s. This does not change the actual port speed, only the metric that the OSPF process on R1 will use to calculate best routes.

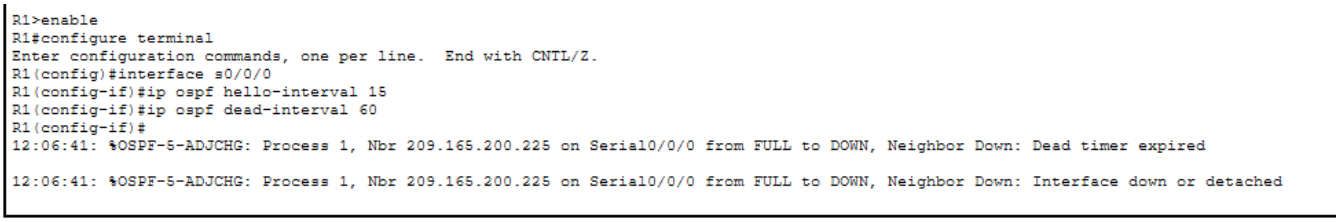
```
R1(config-if)# bandwidth 64
```




รูปที่ 2: การป้อนคำสั่ง bandwidth 64 บน Interface Serial 0/0/0 ของ Router R1

**c.** Trace the path between PC1 and the web server located at `64.100.1.2`. Notice that the path from PC1 to `64.100.1.2` is redirected through R3. OSPF prefers the lower cost path.

```
C:\> tracert 64.100.1.2
Tracing route to 64.100.1.2 over a maximum of 30 hops:
1   1 ms   0 ms   3 ms   172.16.1.1
2   8 ms   1 ms   1 ms   192.168.10.6
3   2 ms   0 ms   2 ms   172.16.3.2
4   2 ms   3 ms   1 ms   209.165.200.226
5   2 ms  11 ms  11 ms   64.100.1.2
Trace complete.
```




รูปที่ 3: ผลการ Tracert จาก Command Prompt หลังปรับค่า Bandwidth (สลับเส้นทางผ่าน R3)

### Part 2: Verify Connectivity

Verify that all PCs can ping the web server and each other.

## Answer Scripts

### Router R1

```
interface s0/0/0
 ip ospf hello-interval 15
 ip ospf dead-interval 60
 bandwidth 64
```

### Router R2

```
interface s0/0/0
 ip ospf hello-interval 15
 ip ospf dead-interval 60
```